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Assignment 3

1. $x[n] = \{1, 2, 1\}$, $n = 0, 1, 2$ $\quad h[n] = \{1, -1, 2\}$, $n = 0, 1, 2$

a). $y[n] = x[n] * h[n]$

$$= \sum_{m=-\infty}^{\infty} x[m]\, h[n-m]$$

$$y[0] = \sum_{m=-\infty}^{\infty} x[m]\, h[-m]$$

$$= x[0]\,h[0] + x[1]\,h[-1] + x[2]\,h[-2] + \cdots$$

$$= \phi + 0 + 0 + \cdots$$

$$= 1$$

$$y[1] = \sum_{m=-\infty}^{\infty} x[m]\, h[1-m]$$

$$= x[0]\,h[1] + x[1]\,h[0] + x[2]\,h[-1] + 0$$

$$= 1 \times -1 + 2 \times 1 + 0$$

$$= 1$$

$$y[2] = \sum_{m=-\infty}^{\infty} x[m]\, h[2-m]$$

$$= x[0]\,h[2] + x[1].h[1] + x[2]\,h[0] + 0$$

$$= 1 \times 2 + 2 \times -1 + 1 \times 1$$

$$= 1$$

$$y[3] = \sum_{m=-\infty}^{\infty} x[m]\, h[3-m]$$

$$= x[0]\,h[3] + x[1]\,h[2] + x[2].h[1] + x[3]\,h[0] + 0$$

$$= 1 \times 0 + 2 \times 2 + 1 \times -1 + 0$$

$$= 3$$

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$$y[4] = \sum_{m=-\infty}^{\infty} x(m)\, h[4-m]$$

$$= x[0].h(4) + x(1).h(3) + x(2).h(2) + x(3).h(1) + 0$$

$$= 0 + 0 + 1\times 2$$

$$= 2$$

$$y[n] = \{1, 1, 1, 3, 2\} \quad n = 0, 1, 2, 3, 4$$

b)

n	$\sum_{m=-\infty}^{\infty} x(m)\, h(n-m)$	$y(n)$
0	1×1	1
1	$1\times -1 + 2\times 1$	1
2	$1\times 2 + 2\times -1 + 1\times 1$	1
3	$1\times 0 + 2\times 2 + 1\times -1$	3
4	1×2	2

2).

a). $h_1(n) = \{1, 1\}, \; n = 0, 1 \qquad h_2(n) = \{1, -1, 1\} \quad n = 0, 1, 2$

$$h_{eq}(n) = h_1(n) * h_2(n)$$

$$= \sum_{k=-\infty}^{\infty} h(k)\, h(n-k)$$

$$h(0) = h_1(0)\, h_2(0)$$

$$= 1$$

$$h_{eq}(n) = \{1, 1, 0, 0\} \quad n = 0, 1, 2, 3$$

$$h(1) = h_1(0)\, h_2(1) + h_1(1).h_2(0)$$

$$= -1 + 1$$

$$= 0$$

$$h(2) = h_1(0)\, h_2(2) + h_1(1)\, h_2(1) + 0$$

$$= 1 + -1$$

$$= 0$$

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duration of $h_{eq}[n] = n_1 + n_2 - 1$

$= 2 + 3 - 1$

$= 4$

EE 325 Digital Signal Processing
Assignment 3: Discrete Time Convolution

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E/21/291

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1 Objective

Verify the calculation of linear convolution for discrete-time sequences using numerical tools.

2 Q1: Linear Convolution

Sequences:

- $x[n] = [1, 2, 1]$
- $h[n] = [1, -1, 2]$

Convolution $y[n] = x[n] * h[n]$**:**

- Length of y: $L_x + L_h - 1 = 3 + 3 - 1 = 5$.
- Result:
  - $y[0] = 1$
  - $y[1] = 1$
  - $y[2] = 1$
  - $y[3] = 3$
  - $y[4] = 2$
- **Verified Output:** $[1, 1, 1, 3, 2]$

3 Q2: Cascade Connection of Systems

Systems:

- $h_1[n] = [1, 1]$
- $h_2[n] = [1, -1, 1]$

Equivalent Impulse Response $h_{eq}[n]$**:** For cascade systems, $h_{eq}[n] = h_1[n] * h_2[n]$.

- Length: $2 + 3 - 1 = 4$.
- Calculation:
  - $h_{eq}[0] = 1$

- $h_{eq}[1] = 0$
- $h_{eq}[2] = 0$
- $h_{eq}[3] = 1$

- **Verified Output:** $[1, 0, 0, 1]$

4 Conclusion

The computational results confirm that numerical convolution matches the analytical expectations for both single-stage and cascade LTI systems.

A Python Source Code

File: `convolution_verification.py`

```python
import numpy as np
import matplotlib.pyplot as plt
import os

def plot_stem(n, y, title, filename):
    plt.figure(figsize=(8, 4))
    plt.stem(n, y, basefmt=" ")
    plt.title(title)
    plt.xlabel('n')
    plt.ylabel('Amplitude')
    plt.grid(True)
    plt.xticks(n)
    path = os.path.join('Convolution', 'plots', filename)
    os.makedirs(os.path.dirname(path), exist_ok=True)
    plt.savefig(path)
    print(f"Saved plot to {path}")
    plt.close()

def main():
    # Q1
    x = np.array([1, 2, 1])
    h = np.array([1, -1, 2])
    nx = np.arange(len(x)) # 0, 1, 2
    nh = np.arange(len(h)) # 0, 1, 2

    y1 = np.convolve(x, h)
    ny1 = np.arange(len(y1)) # 0, 1, 2, 3, 4

    print("Q1 Result y[n]:", y1)
    plot_stem(ny1, y1, 'Q1: Convolution y[n] = x[n] * h[n]', 'Q1_Output.png')

    # Q2
    h1 = np.array([1, 1])
    h2 = np.array([1, -1, 1])

    heq = np.convolve(h1, h2)
    nheq = np.arange(len(heq)) # 0 to 4

    print("Q2 Result heq[n]:", heq)
```

```
    plot_stem(nheq, heq, 'Q2: Cascade Response heq[n]', 'Q2_Output.png')

if __name__ == "__main__":
    main()
```

Listing 1: Discrete-Time Convolution Verification – Assignment 3